INS3441

HIGHWAY ENGINEERING

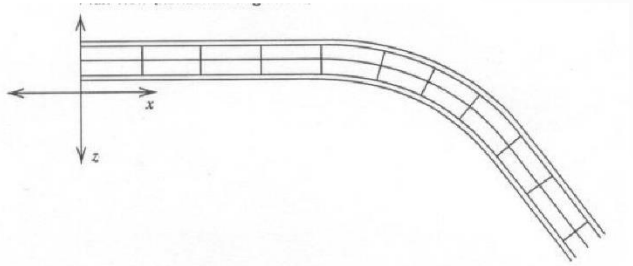
Prof. Halit ÖZEN, PhD.

COURSE SYLLABUS

WEEK	SUBJECT	DATE	
1	INTRODUCTION BASIC DEFINITIONS (TERMS)		
2	ROAD USER CHARECTERISTICS VEHICLE MOVEMENTS		
3	TRAFFIC FLOW AND CAPACITY ROADWAY CASSIFICATION AND GEOMETRIC STANDARDS		
4	HIGHWAY ROUTE STUDY **NATIONAL HOLIDAY**		
5	HORIZANTAL CURVE AND SUPERELEVATION (Project sheet distribution) HORIZANTAL CURVE AND SUPERELEVATION		
6	**APPLICATION (VEHICLE MOVEMENTS)** **ASSIGNMENT - APPLICATION (TRAFFIC)**		
7	**APPLICATION (HORIZANTAL CURVE)** **ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)**		
8	**MID TERM EXAM**		
9	TRANSITION CURVES- ASSIGNMENT (HAND IN) TRANSITION CURVES – ASSIGNMENT (RETURN)		
10	VERTICAL ALIGNMENT AND VERTICAL CURVE VERTICAL ALIGNMENT AND VERTICAL CURVE		
11	**APPLICATION (TRANSITION CURVES)** **ASSIGNMENT**		
12	**APPLICATION (VERTICAL CURVE)** **APPLICATION (VERTICAL AND HORIZANTAL CURVE)**		
13	PLANNING OF INTERSECTION AND URBAN ROADS PROJECTS DELIVERY DUE DATE (December 30, 2020)		
14	DRAINAGE IN HIGHWAY MAKEUP EXAM		

Components of The Alignment

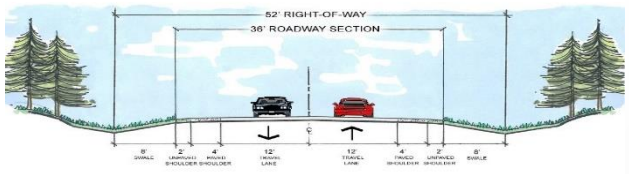
Horizontal Alignment



Vertical Alignment



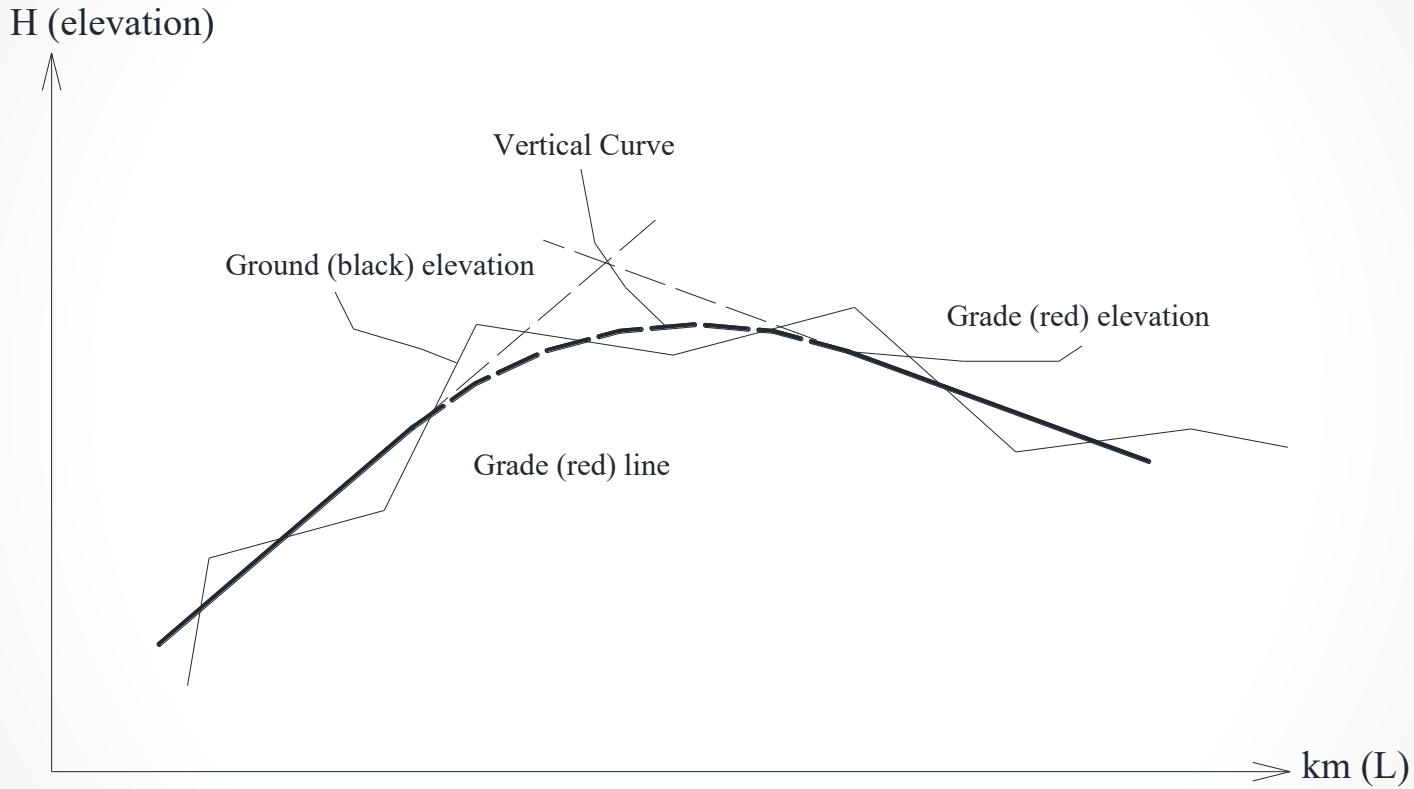
Cross-section



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PROFILE DESIGN AND VERTICAL CURVES

The section taken through the highway axis is named as the **profile**.



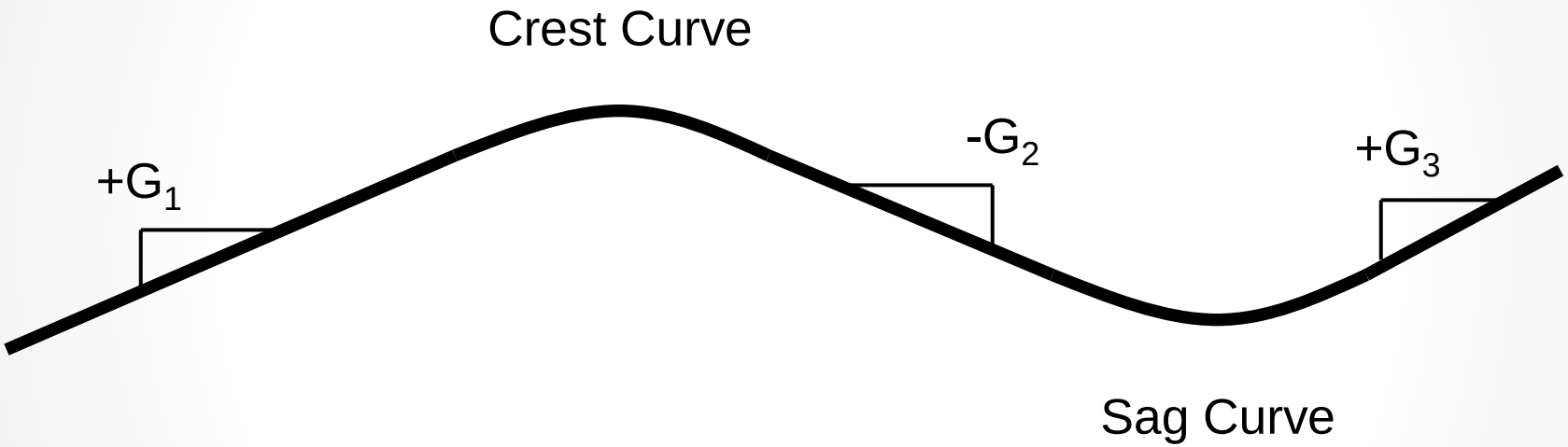
The longitudinal grade given through the highway axis is shortly named as the **grade**.

VERTICAL CURVES

Tangents and Curves

Like the horizontal alignment, the vertical alignment is made up of tangent and curves

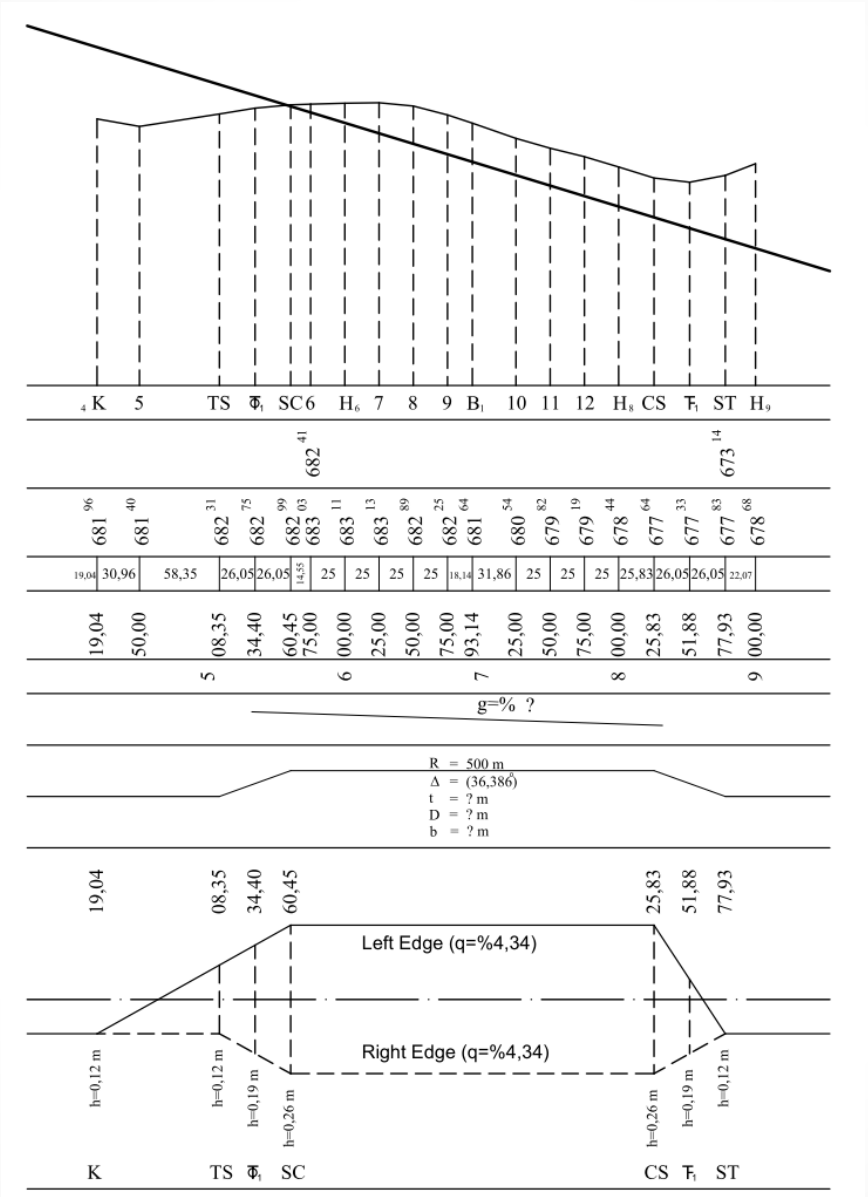
In this case the curve is a parabolic curve rather than a circular or spiral curve



If the grade of the red line of the profile is positive (+), this section of the highway is an ascending (upgrade) section, else if the grade is negative (-), this section of the highway is a descending (downgrade) section.

G is algebric difference in grades. $\mathbf{G = g_1 - g_2}$

PROFILE


K 5 TS SC 6 7 8 9 10 11 12 CS ST
682 673
681 681 682 682 682 683 683 683 682 682 681 680 679 679 678 677 677 677 678
19,04 30,96 58,35 26,05 26,05 14,55 25 25 25 25 18,14 31,86 25 25 25 25,83 26,05 26,05 22,07
19,04 50,00 08,35 34,40 60,45 75,00 00,00 25,00 50,00 75,00 93,14 25,00 50,00 75,00 00,00 25,83 51,88 77,93 00,00
5 6 7 8 9
g=% ?
R = 500 m
Δ = (36,386)
t = ? m
D = ? m
b = ? m
19,04 08,35 34,40 60,45 25,83 51,88 77,93
Left Edge (q=%4,34)
Right Edge (q=%4,34)
h=0,12 m h=0,12 m h=0,19 m h=0,26 m h=0,26 m h=0,19 m h=0,12 m
K TS SC CS ST

Maximum Grade



Maximum and Minimum Grade

One important design consideration is the determination of the maximum and minimum grade that can be allowed on the tangent section

The minimum grade used is typically 0.5%

The maximum grade is generally a function of the

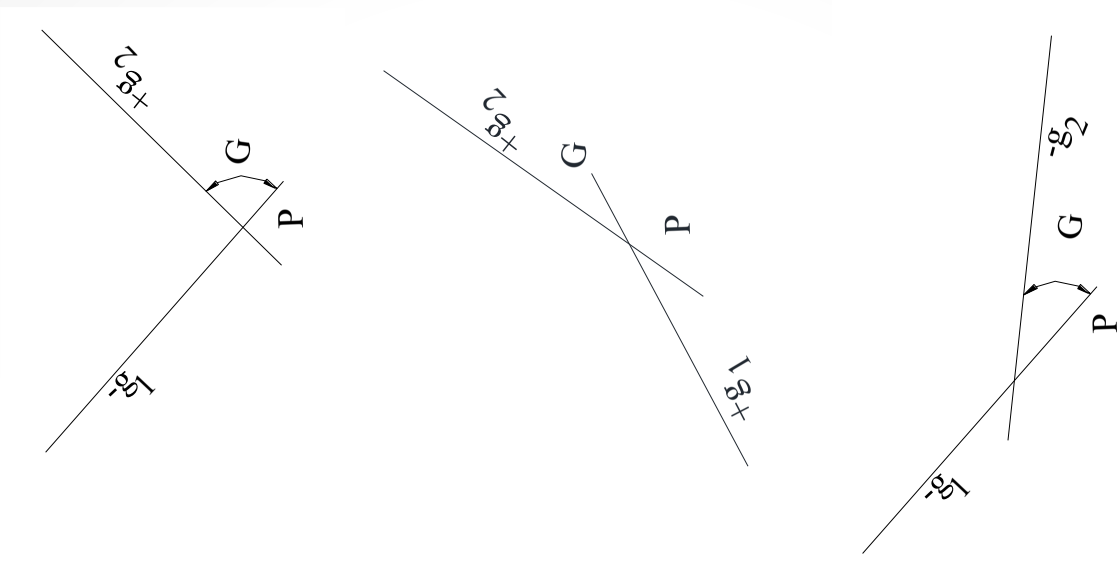
- Design Speed
- Terrain (Level, Rolling, Mountainous)

On high-speed facilities such as freeways the maximum grade is generally kept to 5% where the terrain allows (3% is desirable since anything larger starts to affect the operations of trucks)

At 50 km/h design speed the acceptable maximum is in the range of 7 to 12 %

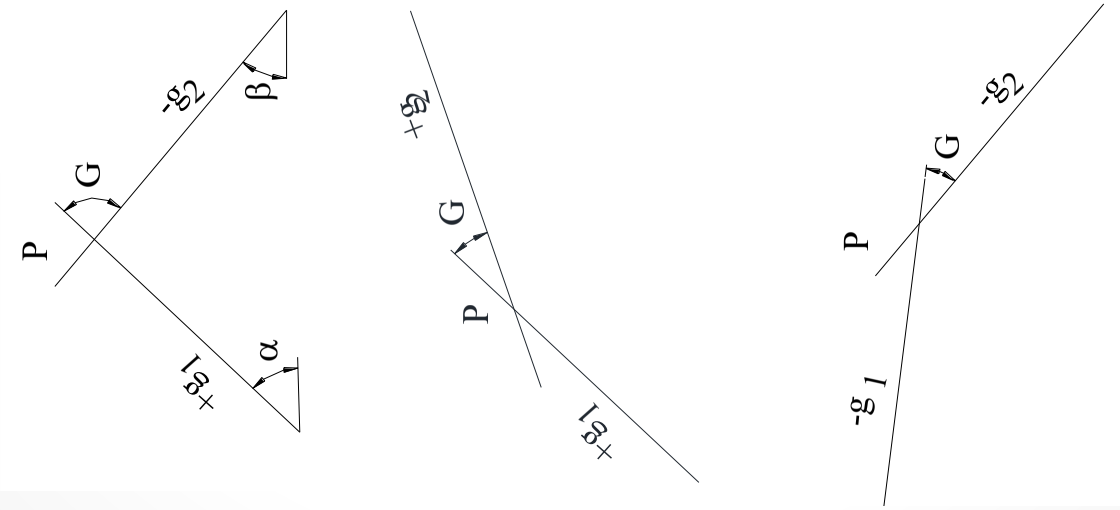
Sag Curve

the algebraic difference is negative (-)



Crest Curve

the algebraic difference is positive (+)



VERTICAL CURVES

Sight distance on vertical curves

On **undivided highways,** the most inappropriate condition arises when a driver overtakes a vehicle in front according to the possibility of a vehicle approaching from the opposite direction.

$$S = L_s = \frac{(d_1 + d_2)}{(v_1 - v_2)} * (v_1 + v_3)$$

As there isn't any vehicle approaching from the opposite direction on a divided highway, usually the necessary brake safety distance to avoid crashing with an obstacle is considered.

$$S = l_{fe} = 0{,}278.V * t_r + 0{,}00394 * \frac{V^2}{(f \pm s)}$$

Comfort at vertical curves

While the vehicles are moving on the vertical curve, they are affected by a vertical acceleration that creates discomfort just like the horizontal curves.

This value is around 0,5 m/sec^2

$$F = m * \frac{V^2}{R} \Longrightarrow \boxed{a = \frac{v^2}{R}}$$

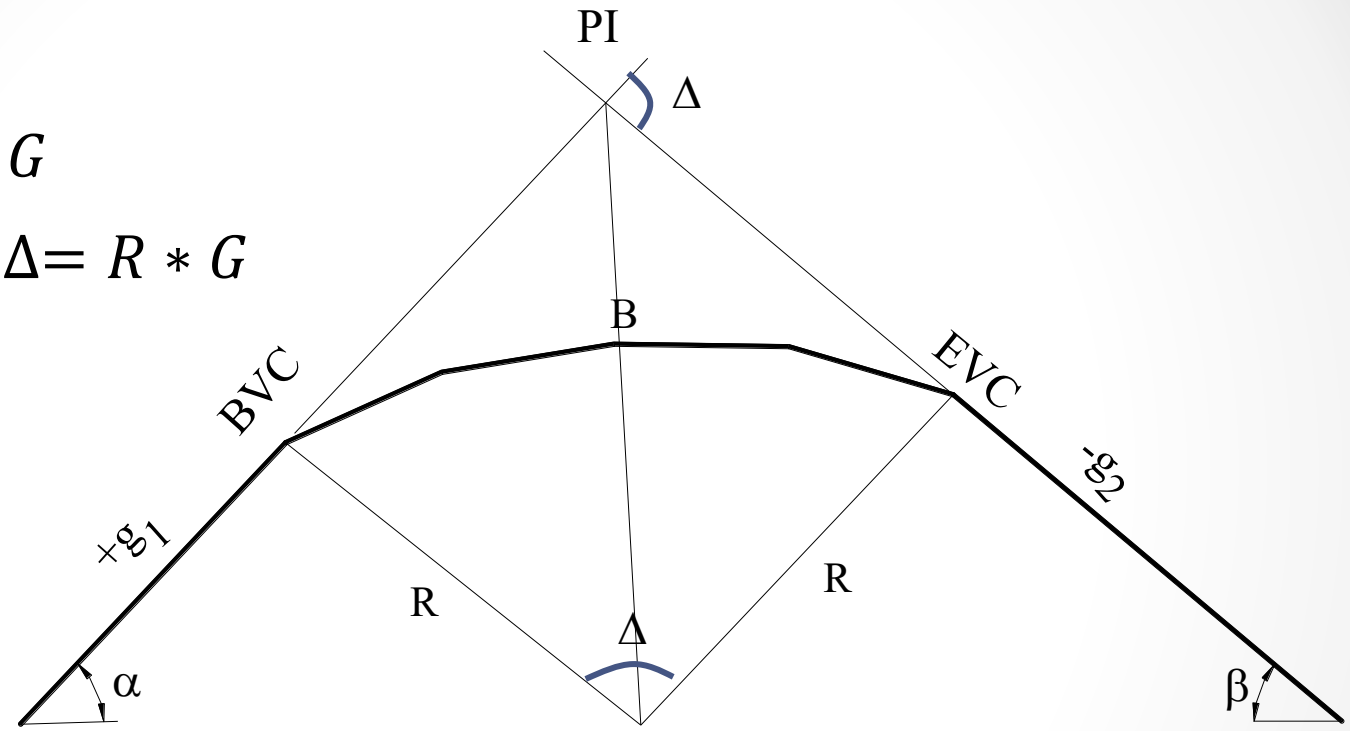
VERTICAL CURVES

$$g_1 = \tan\alpha$$

$$g_2 = \tan\beta$$

$$\Delta \cong \tan\Delta = G$$

$$L = R * \tan\Delta = R * G$$



$$L = \frac{V^2}{13 * a} * G$$

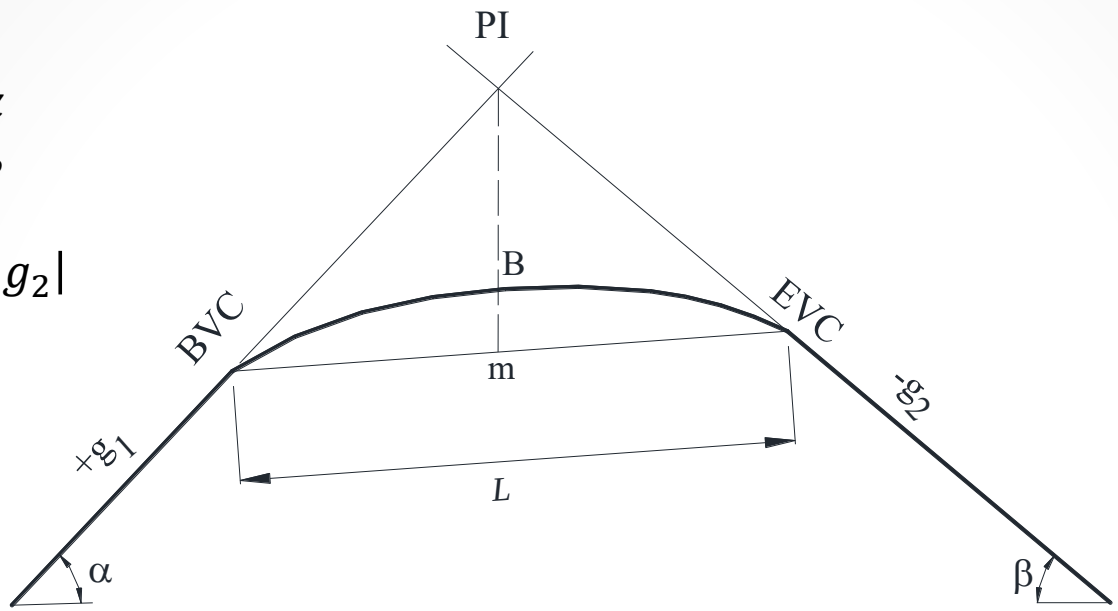
VERTICAL CURVES

Parabolic vertical curves

$g_1 = \tan\alpha$

$g_2 = \tan\beta$

$G = |g_1 - g_2|$



$\overline{BVCPIEVC} = BV\widehat{CBE}VC = \overline{BVCmEVC}$

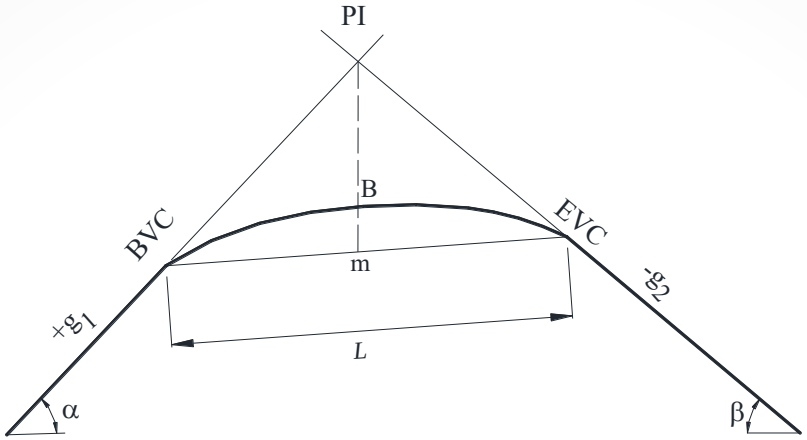
$y = k * x^2$

$\overline{PB} = \dfrac{\overline{PM}}{2}$

$\overline{PB} = e$

$y = k * x^2 \qquad \overline{PB} = e \qquad e = k * x_1^2 \qquad e = k * x_2^2 \qquad x_1 = x_2 = \dfrac{L}{2}$

VERTICAL CURVES



$$y = k * x^2 \qquad \overline{PB} = e \qquad y' = 2 * k * x \qquad y'' = 2 * k = \frac{G}{L}$$

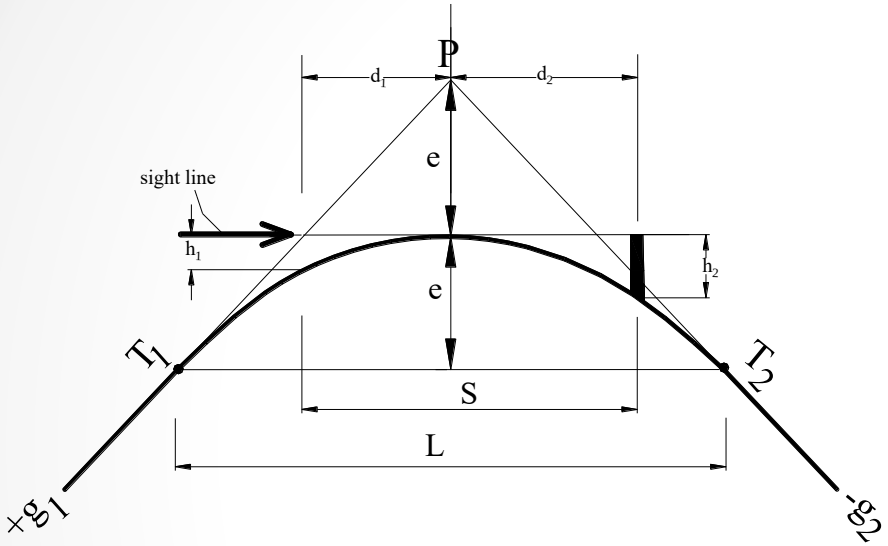
$$\boxed{k = \frac{G}{2 * L}}$$

$$\boxed{y = \frac{G}{2 * L} * x^2}$$

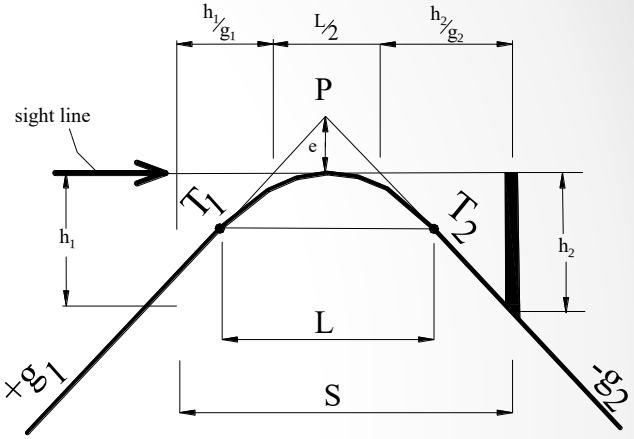
$$x = L/2 \qquad y = e \quad \Longrightarrow \quad \boxed{e = \frac{G * L}{8}}$$

VERTICAL CURVES

Calculations of parabolic vertical curves

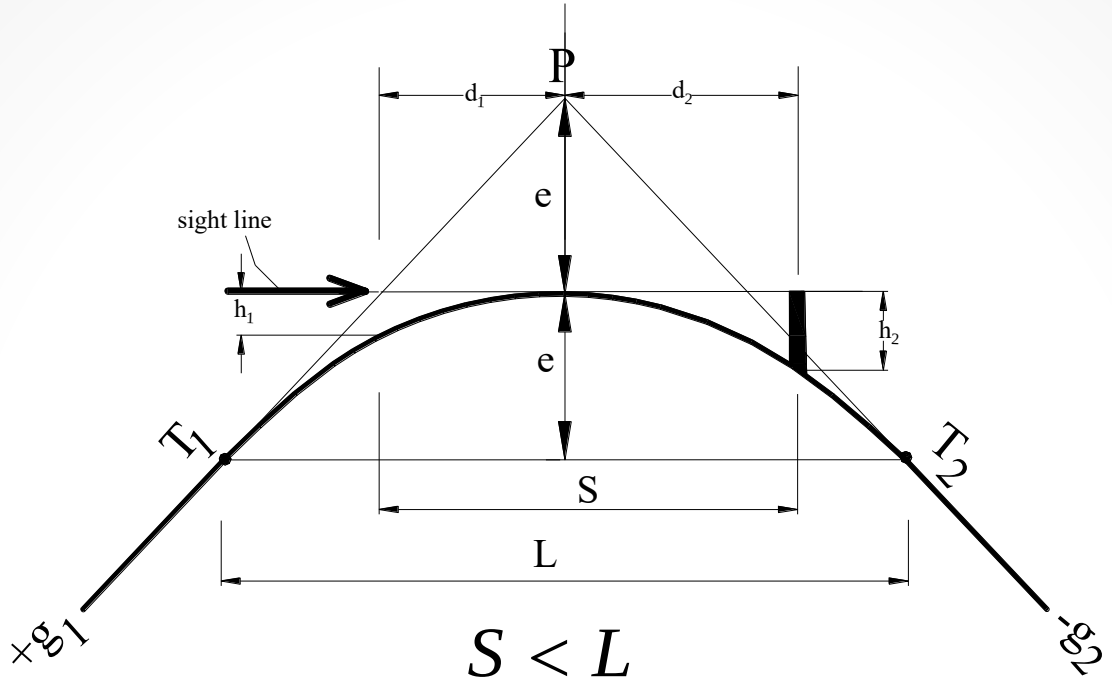


$S < L$



$S > L$

VERTICAL CURVES



$$S < L$$

$$h_1 = k * d_1^2$$

$$d_1 = \sqrt{\frac{h_1}{k}} = \sqrt{\frac{h_1 * L^2}{4 * e}}$$

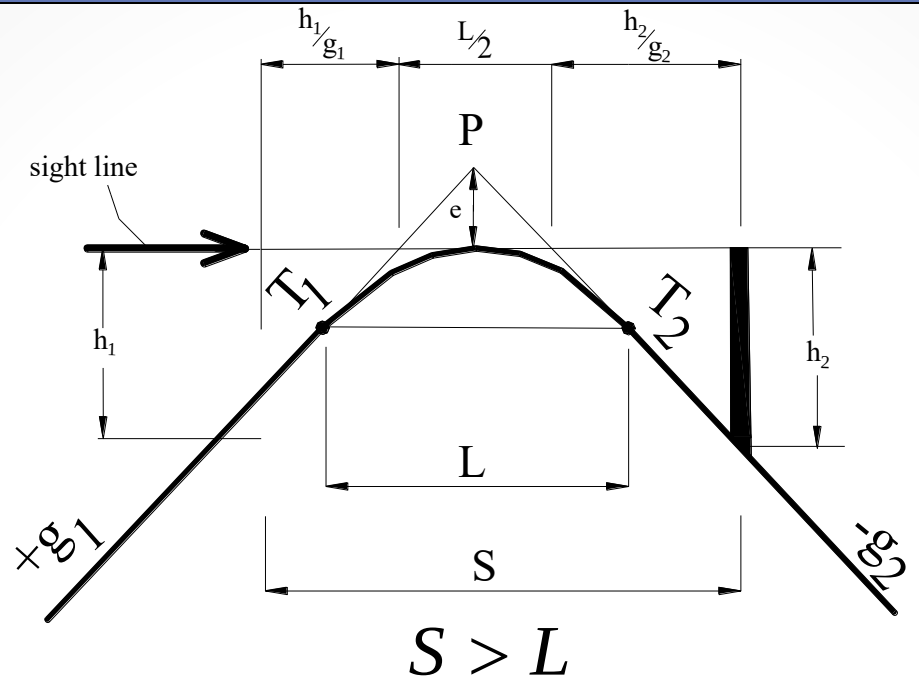
$$d_2 = \sqrt{\frac{h_2}{k}} = \sqrt{\frac{h_2 * L^2}{4 * e}}$$

$$d_1 + d_2 = \sqrt{\frac{h_1 * L^2}{4 * \frac{L * G}{8}}} + \sqrt{\frac{h_2 * L^2}{4 * \frac{L * G}{8}}}$$

$$d_1 + d_2 = \frac{L}{G} * \left(\sqrt{2 * h_1} + \sqrt{2 * h_2}\right)^2 = S$$

$$\boxed{L = \frac{G * S^2}{\left(\sqrt{2 * h_1} + \sqrt{2 * h_2}\right)^2}}$$

VERTICAL CURVES



$$S = \frac{L}{2} + \frac{h_1}{g_1} + \frac{h_2}{g_2} \qquad S = \frac{L}{2} + \frac{h_1}{g_1} + \frac{h_2}{G - g_1}$$

$$S = \frac{L}{2} + \frac{h_1}{\left(\frac{G * \sqrt{h_1 * h_2} - h_1 * G}{h_2 - h_1}\right)} + \frac{h_2}{\left(\frac{G * \sqrt{h_1 * h_2} - h_1 * G}{h_2 - h_1}\right)}$$

Minumun sight distance occurs if $ds/dg_1 = 0$. Therefore;

$$\frac{dS}{dg_1} = -\frac{h_1}{g_1} + \frac{h_2}{(G - g_1)^2} = 0 \qquad g_1 = \frac{G * \sqrt{h_1 * h_2} - h_1 * G}{h_2 - h_1}$$

$$\boxed{L = 2 * S - \frac{2 * \left(\sqrt{h_1} + \sqrt{h_2}\right)^2}{G}}$$

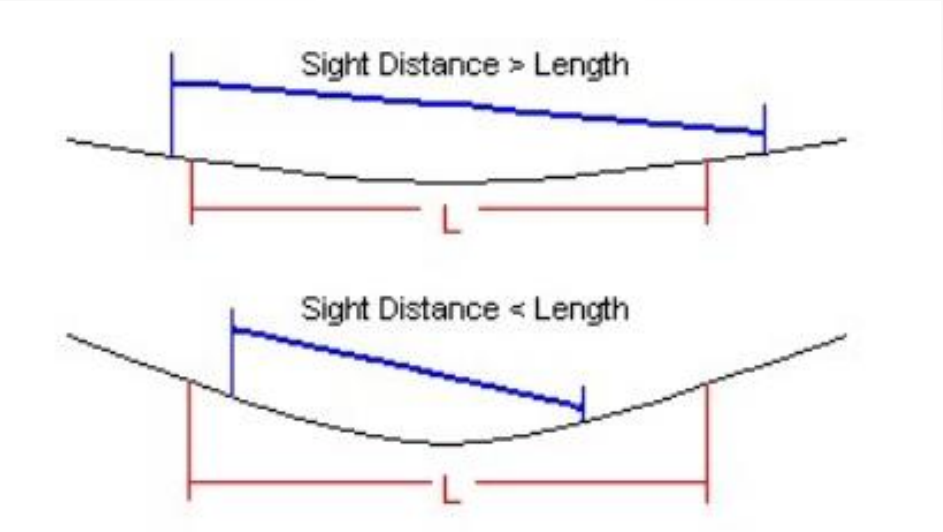
VERTICAL CURVES

If h_1 =1,37 m h2=15 cm, equations are;

$$S < L \quad \Rightarrow \quad L = \frac{G * S^2}{10}$$

$$S > L \quad \Rightarrow \quad L = 2 * S - \frac{10}{G}$$

VERTICAL CURVES



Sight distance based on head light;

$$S < L \quad \Rightarrow \quad L = \frac{G * S^2}{2 * (h + S * \tan\alpha)}$$

$$S > L \quad \Rightarrow \quad L = 2 * S - \frac{2 * (h + S * \tan\alpha)}{G}$$

Sight distance based on obstacle at top like bridges;

$$S < L \quad \Rightarrow \quad L = \frac{G * S^2}{8 * \left(H - \frac{h_1 + h_2}{2}\right)}$$

$$S > L \quad \Rightarrow \quad L = 2 * S - \frac{8}{G} * \left(H - \frac{h_1 + h_2}{2}\right)$$

VERTICAL CURVES

Circular vertical curves

$$g_1 = \tan\alpha$$

$$g_2 = \tan\beta$$

$$G = |g_1 - g_2|$$

$$\text{tg}\,\gamma = \text{tg}(\alpha + \beta)$$

$$\text{tg}\,\gamma = \frac{\text{tg}\alpha + \text{tg}\beta}{1 - \text{tg}\alpha\ \text{tg}\beta}$$

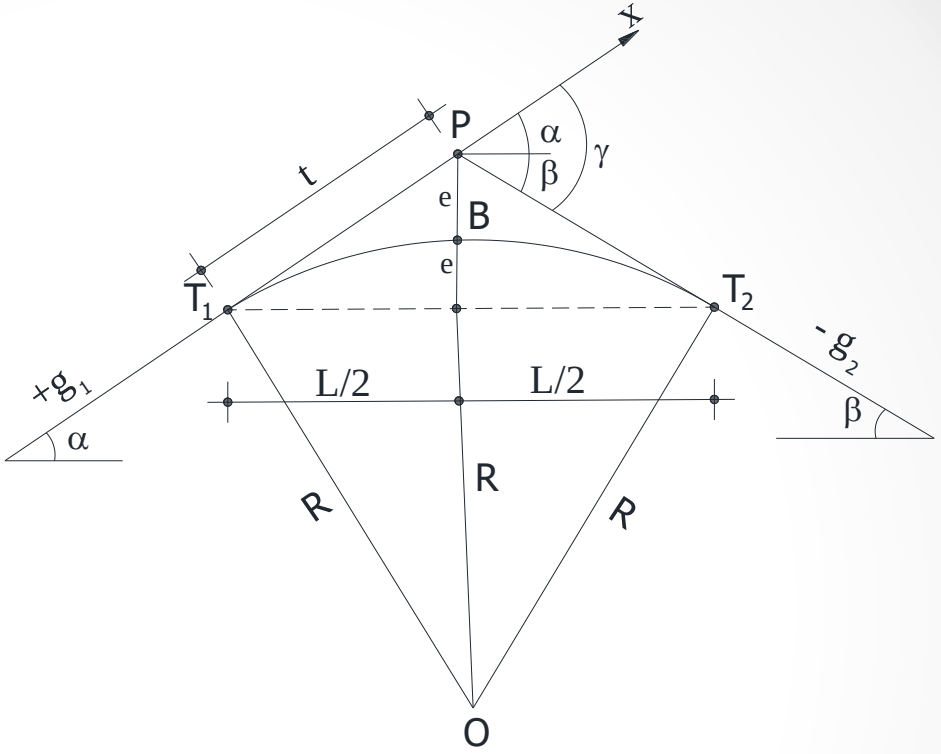
neglected

$$\text{tg}\,\gamma \cong \text{tg}\alpha + \text{tg}\beta = g_1 + g_2$$

$$\gamma \cong \text{tg}\,\gamma = G$$

$$t = R * tg\left(\frac{\gamma}{2}\right) = R * \frac{G}{2}$$

$$2t = L = R * G$$



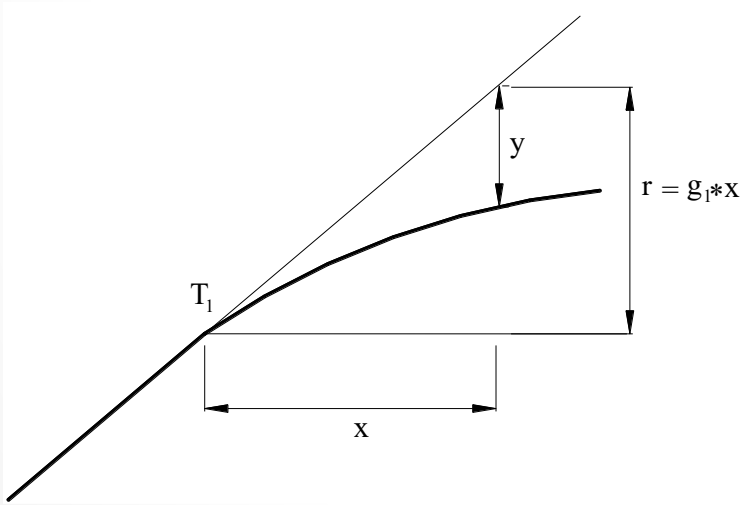
$$L = R * G$$

L_{min}= 120 m	R_{min}= 750 m (Crest)
L_{min}= 80 m	R_{min}= 500 m (Sag)

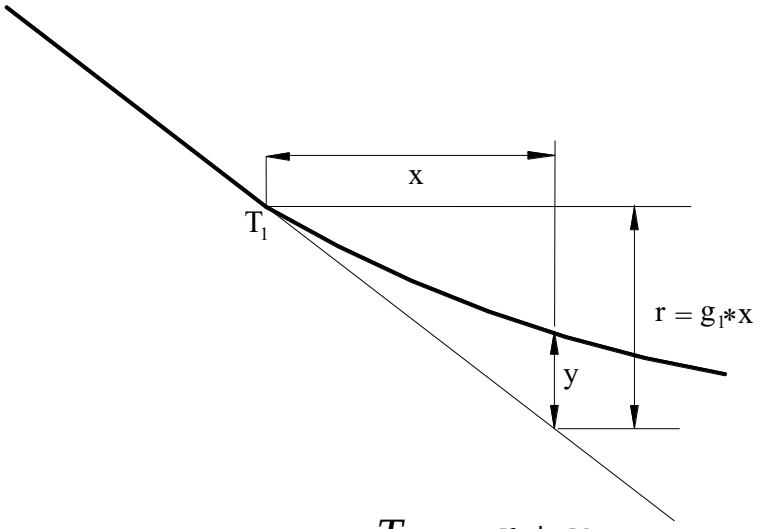
Vertical Curve Type and the Terrain Type	R_{min} (m) according to the highway type			
	I	II	III	IV
Crest Vertical Curve				
Flat	25.000	15.000	10.000	5.000
Undulating	15.000	10.000	5.000	2.500
Mountainous	5.000	2.500	1.500	1.000
Sag Vertical Curve				
Flat	8.000	5.000	3.000	2.000
Undulating	5.000	3.000	2.000	1.500
Mountainous	2.000	1.500	1.200	1.000

VERTICAL CURVES

Cross-section	km	x (m)	$r = g_1 * x$	$PVC_{elv} + r$	$y = k * x^2$	$PVC_{elv} + r \pm y$
T_1						PVC_{elv}
B		x=t		P_{elv}	y=e	B_{elv}
T_2		x=L				PVT_{elv}



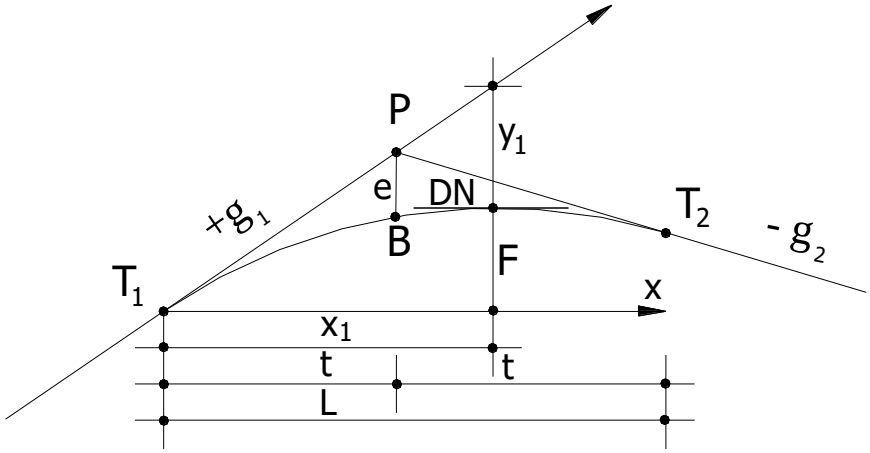
$T_{1kot} + r - y$



$T_{1kot} - r + y$

Calculation of turning point at vertical curves

The point having the maximum or minimum elevation on the vertical curve is named as the **turning point (TP-DN)**.



$$TP(DN)_{km} = T1_{km} + \frac{g_1 \quad L}{G}$$

$$TP(DN)_{elv} = T1_{elv} + \frac{g_1^2 \quad L}{2G}$$

Question: On a road section with a Design Speed of 80 km/h, there are two intersecting vertical allignments at a point P. The slopes are g_1= 0.055 and g_2= - 0.025 respectively. KM value of the point P is given as (2+400) m and its elevation is 185 m. It is planned to place a circular vertical curve with a Radius of R =5000 m between those vertical allignments. Calculate the Km values and elevations of the beginning and end points, bisector, and inflection point of the vertical curve.

$$G = g_1 - g_2 = 0.055 - (-0.025) = 0.08$$

$$t = \frac{G * R}{2} = \frac{0.08 * 5000}{2} = 200 \ \ m \quad \rightarrow \quad L = 2 * t = 2 * 200 = 400 \ \ m$$

$$e = \frac{G * L}{8} = \frac{0.08 * 400}{8} = 4.0 \ \ m$$

$$y = K * x^2 \qquad K = \frac{1}{2 * R} = \frac{1}{2 * 5000} = 0.0001 \quad \rightarrow \quad y = 0.0001 * x^2$$

Cross-section	km	x (m)	$r = g_1 * x$	$PVC_{elv} + r$	$y = k * x^2$	$PVC_{elv} + r \pm y$
T_1						PVC_{elv}
B		x=t		P_{elv}	y=e	B_{elv}
T_2		x=L				PVT_{elv}

KM Calculation	Elevation Calculation
$T_{1km} = P_{km} - t = 2400 - 200 = 2 + 200$	$T_{1kot} = P_{kot} - g_1 * t = 185 - 0.055 * 200 = 174$ m
$T_{2km} = P_{km} + t = 2400 + 200 = 2 + 600$	$T_{2kot} = P_{kot} + g_2 * t = 185 - 0.025 * 200 = 180$ m
$B_{km} = P_{km} = 2 + 400$	$B_{kot} = P_{kot} - e = 185 - 4 = 181$ m
$DN_{km} = T_{1km} + \frac{g_1 * L}{G}$	$DN_{kot} = T_{1kot} + \frac{(g_1)^2 * L}{2 * G}$
$DN_{km} = 2200 + \frac{0.055 * 400}{0.08} = 2 + 475$	$DN_{kot} = 174 + \frac{(0.055)^2 * 400}{2 * 0.08} = 181.56$ m